Multi-Agent LLM System for Real-Time Portfolio Risk Management with Causal Inference

Capstone Project Proposal · FSE 570 · M.S. Data Science, Analytics & Engineering
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1. Problem Statement

What are we trying to do, how is it done today, what are its limits, and who benefits? *Heilmeier Q1–Q3, Q5*

Most portfolio risk today is measured using Value at Risk and Conditional VaR. Both are built from a covariance matrix of asset returns, so they describe how assets have moved together in the past. The weakness is that those relationships are not stable. They tend to change during the market shifts the models are supposed to warn about, so a portfolio that looked diversified can stop being diversified very quickly.

Two recent episodes show what this costs. Between 19 February and 23 March 2020 the S&P 500 fell about 34%, and the VIX closed at 82.69, its highest close on record. In March 2023 Silicon Valley Bank, which held roughly $209 billion in assets at the end of 2022, was closed by regulators within days of a deposit run. In both cases standard risk measures reacted after the damage rather than before it.

A useful system needs three things: some sense of direction rather than just co-movement, so a user can see which part of the market is passing stress to the rest; regular updating as new data arrives; and an explanation attached to each alert, since risk teams tend to ignore signals they cannot check.

The users would be risk managers at asset managers, insurers and pension funds, who decide when to hedge or cut exposure; behind them are savers whose retirement money absorbs the losses. Demand is real. The Office of Financial Research and the Chicago Fed both publish financial stress indices, which also gives us a public benchmark to measure against.

2. Data Sources

What data does this take, and can we actually get it?

We plan to use three public datasets.

1. Market prices, from Yahoo Finance via the yfinance library. Daily price and volume data for roughly 20–30 exchange-traded funds covering equity sectors, Treasuries, corporate credit, commodities and the dollar, from around 2005 onwards. That gives roughly 5,000 daily observations per series.

2. Macroeconomic indicators, from FRED (Federal Reserve Bank of St. Louis). We plan to pull the VIX, credit spreads, the Treasury term spread, unemployment, the Chicago Fed financial conditions index and similar series. These arrive at different frequencies, from daily to monthly.

3. News text, from FNSPID (published at KDD 2024), possibly supplemented with GDELT. FNSPID contains millions of news records already matched to trading dates. This dataset feeds the explanation part of the system only.

Cleaning and integration. Most of the work is getting these onto one common daily timeline. Prices need split and dividend adjustment, and not every fund exists for the whole period, so the panel starts ragged. Macro series are revised after release, so where we can we will use FRED's archival version (ALFRED), which returns the value actually available at the time. News needs deduplication and timestamp cleanup, and we will check the series for stationarity before modelling.

Access and privacy. All three are free and public; FRED needs only a free API key. We are not using proprietary data, and none of these datasets contains personal information, since they cover securities, economic aggregates and published news about public companies. The main practical risk is rate limits or a source changing format, so we will store our own copy early and work from that.

3. Methodology

What is new in the approach, and how will we know whether it works? *Heilmeier Q4, Q9*

The flow is: build the dataset, estimate how the variables relate to each other over rolling windows, turn changes in those relationships into a feature, then test whether that feature predicts market stress.

Causal discovery. We plan to use VarLiNGAM, a method for estimating a directed structural vector autoregression (Hyvärinen et al., 2010). A standard VAR shows which series lead others but cannot easily assign direction to same-day effects, and VarLiNGAM uses the non-normal shape of return distributions to help with that. We will re-estimate on rolling windows and run ordinary Granger causality tests as a simpler comparison. These are directed relationships under assumptions financial data only partly satisfies, so we will describe the output as directed dependence rather than proven causality.

Feature. From one window to the next we measure how much the estimated structure changes, meaning how many links appear or disappear and how much their strengths move. That measure is the signal we are testing.

Target. We will use a Markov regime-switching model (Hamilton, 1989) on realised volatility to label calm and stressed periods, rather than choosing a cutoff by hand.

Models. A logistic regression predicting a shift into a stressed regime, fit in a Bayesian form so we get a range for the estimated warning time rather than a single number, plus a gradient boosting model as a comparison for nonlinear effects.

Validation. All splits are by time. We will not use random train/test splits, since they let future information into training. We train on earlier data, test on the following period, move forward and repeat across the sample, tuning only inside the training portion. Baselines: a GARCH-based correlation model, rolling-correlation VaR, the VIX on its own, and the Chicago Fed index. We will report how far ahead of a stressed period the signal moves, plus precision, recall and false-alarm rate. The 2020 and 2023 episodes are shown as examples, but the main results come from the full rolling test, since two events are not enough to conclude much.

Explanation layer. If the earlier stages work and time allows, we will add a small set of LLM agents (one reading news, one reading macro releases, one querying the estimated graph) coordinated to write a short explanation for each alert. Worth saying up front: a language model trained after 2023 already knows how COVID and SVB played out, so we will not claim it predicted them. Anything we say about early detection comes from the statistical side, which has no access to future data. If we get that far we would also wrap the output in a simple service and dashboard so alerts can be viewed as they update, but that is the last thing we would build and the first we would drop.

4. Project Management

What are the risks, how long will it take, and what will it cost? *Heilmeier Q6–Q8*

Weeks	Focus	What we expect to have done
1–3	Data	Data pulled, cleaned and aligned; a stored copy we can work from
4–6	Model	Rolling estimates running over the full period; instability feature computed
7–8	Testing	First real comparison against the baselines; decision point on whether to continue as planned
9–11	Explanation layer	LLM agents producing written alerts; basic interface
12–14	Write-up	Final report, visualisations and presentation

Risk 1: the signal may not work. The instability measure may simply not beat the VIX or the Chicago Fed index. We will start comparing against those baselines around Week 5 on a small held-out slice rather than waiting until the end. Week 8 is our decision point; if it is not working we will report that and spend the remaining weeks explaining why.

Risk 2: the estimated relationships may be unstable. Financial data is not stationary, so the structure we estimate could change between nearby windows for reasons unrelated to market stress. We will check this by comparing results across overlapping windows and across both methods, and if the estimates are too noisy we will lengthen the window or average across methods.

Risk 3: the explanation layer may take longer than expected. Agent workflows are the least predictable part of the project. We are treating this stage as optional from the start: data, model and testing are a complete project on their own, and we will build the agent layer only if we are on schedule after Week 8. There is also a risk the agents write plausible but unsupported explanations, so we will restrict them to the retrieved documents and the estimated graph and check a sample by hand. Costs are small, since the data is free and compute needs are modest.

5. Expected Outcome

If it works, what difference does it make? *Heilmeier Q5*

We hope to end up with a monitoring tool rather than a trading signal: something that flags when the structure of the market is changing, shows which relationships are moving, and writes a short explanation a person can check. The contribution would be showing whether directional relationships give any earlier warning than the correlation-based measures already in use. If the answer is no, that is still a result we can report and explain.

References. Hyvärinen et al. (2010), JMLR 11; Hamilton (1989), Econometrica 57(2); Engle (2002), JBES 20(3); Dong, Fan & Peng (2024), FNSPID, KDD '24. Figures: FRED series SP500 and VIXCLS; Cboe VIX historical data; FDIC Press Release PR-23-016, 10 March 2023.